

Project #5: Transportation Project #1 - *Autonomous Vehicle Interactions at an Intersection*

Client: Nebraska Department of Transportation (NDOT)

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Project Goals:

The objective of this project is to analyze the “Foggy Bottom” subset of the “Third Generation Simulation” dataset to determine if Autonomous Vehicles (AVs) are generally safer and more predictable than other agents. This is to develop an understanding on how AVs interact with one another and interact with traditional road users. Using the 200 second data set, the team will develop a Python code to aid in analysis. Analysis includes analyzing localized behaviors near the intersection, including the speed profiles and acceleration/deceleration patterns. Additionally, comparisons between AV’s, non AV’s, and pedestrians will be made.

This project aims to:

- Determine whether AVs exhibit safer and more predictable behavior compared to other agents.

- Identify measurable differences in acceleration, speed, and interaction patterns.
- Provide data-driven insights to support transportation decision-making.

Project Tasks:

The engineering team will be able to complete the following tasks:

- **Task 1: Dataset Understanding and Preparation**
 - Review data structure and variables using provided files.
 - Identify key variables such as, position, speed, acceleration, and agent type.
 - Clean and process data in a way that removes errors, handles missing values, and filtering relevant data.
 - Define intersection boundaries using provided polygon tools.
- **Task 2: Behavioral Analysis**
 - Acceleration and deceleration pattern analysis within intersection zones.
 - Evaluate speed profiles along selected lanes.
 - Calculate average speeds inside the intersection.
 - Count deceleration events as indicators of disruptions
 - Compare results across AVs, human-driven vehicles, and pedestrians.
- **Task 3: Visualization of Results**
 - Create plots such as, speed vs. time, position vs. time, acceleration distributions.
 - Generate summary tables comparing agent behavior.
 - Develop visual representation of agent movements.
- **Task 4: Interpretation**
 - Interpret results in the context of safety and predictability.

Project Deliverables:

The engineering team will provide several key deliverables to support the client's evaluation of autonomous vehicle behavior. First, a dataset description will be prepared to describe the structure of the TGSIM data, such as the variables used, organization of the data, and any processing done prior to analysis. Additionally, initial analysis results will be prepared in the form of tables and graphs that show emerging trends in agent behavior. The main deliverable from this project will be the final report containing the complete analysis. The final report will particularly focus on the comparison between the behavior of autonomous cars and humans driving cars and walking on the roads. In order to help visualize the interactions of the agents, a diagram that shows their movement in the intersection will be supplied, either as a graph or in the form of an animation that demonstrates how the agents interact through movement over time. Lastly, all analysis will be backed up by comprehensive Python code and all will be turned in May 8th, 2026 at 5 pm,